

Frobenius Separation of Massive and Massless Sectors in $GF(27)^*$

Algebraic Structure of the $\tilde{T} \cdot m = 1$ Transition
and Connection to 8 Gluons

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29 August 2026

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Frobenius Separation of Massive and Massless Sectors in $GF(27)^*$

Abstract

The multiplicative group $GF(27)^* \cong \mathbb{Z}_{26}$ decomposes under the Frobenius automorphism $\phi : x \mapsto x^3$ into exactly two types: two fixed points $\{+1, -1\}$ and eight three-element orbits. Via the decomposition $\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}$ (CRT), four $\mathbb{Z}_{13}$ three-orbits times two $\mathbb{Z}_2$ parities yield the eight gluons of the adjoint $SU(3)$ representation; the fixed field $GF(3)$ of the Frobenius supplies the $U(1)$ sector (photon). The fixed points $\{+1, -1\}$ correspond to the massive $\mathbb{Z}_2$ sector (particle/antiparticle). All results are algebraically forced. Status: **[K]**.

Starting Point

In Doc. 338

$$\left(\frac{m_\mu}{m_e}\right)^2 = 43200 = |GF(9)^*|^2 \cdot 5^2 \cdot |GF(27)| = 64 \cdot 25 \cdot 27$$

is established as an algebraically exact relation **[B]**. The fundamental relation $\tilde{T} \cdot m = 1$ describes a duality: massive sector ($m > 0$, compact torus S_m^1) and massless sector ($m \rightarrow 0$, $\tilde{T} \rightarrow \infty$). In Doc. 285 the $3'$ extension is identified as a kinetic (massless) window. This note asks: what algebraic structure separates both sectors within $GF(3) \subset GF(9) \subset GF(27)$?

Decomposition of $\mathbb{Z}_{26}$ under the Frobenius

Basic structure **[B]**

$GF(27)^* \cong \mathbb{Z}_{26}$, cyclic of order $3^3 - 1 = 26 = 2 \cdot 13$. Since $\gcd(2, 13) = 1$:

$$\mathbb{Z}_{26} \cong \mathbb{Z}_2 \times \mathbb{Z}_{13}.$$

The Frobenius $\phi : x \mapsto x^3$ acts:

- On $\mathbb{Z}_2$: $3 \equiv 1 \pmod{2}$ — ϕ is *trivial*.
- On $\mathbb{Z}_{13}$: $\text{ord}(3, \mathbb{Z}_{13}) = 3$ (since $3^3 = 27 \equiv 1 \pmod{13}$) — ϕ has order 3.

Orbits of ϕ on $\mathbb{Z}_{13}$ **[B]**

Type	Elements
Fixed point	$\{0\}$
Three-orbit 1	$\{1, 3, 9\}$
Three-orbit 2	$\{2, 5, 6\}$
Three-orbit 3	$\{4, 10, 12\}$
Three-orbit 4	$\{7, 8, 11\}$

Complete orbit table in $\mathbb{Z}_{26}$ **[B]**

$\mathbb{Z}_2$ parity	$\mathbb{Z}_{13}$ orbit	$\mathbb{Z}_{26}$ orbit
0	$\{0\}$	$\{0\}$ (fixed point)
1	$\{0\}$	$\{13\}$ (fixed point)
0	$\{1, 3, 9\}$	$\{14, 16, 22\}$
1	$\{1, 3, 9\}$	$\{1, 3, 9\}$
0	$\{2, 5, 6\}$	$\{2, 6, 18\}$
1	$\{2, 5, 6\}$	$\{5, 15, 19\}$
0	$\{4, 10, 12\}$	$\{4, 10, 12\}$
1	$\{4, 10, 12\}$	$\{17, 23, 25\}$
0	$\{7, 8, 11\}$	$\{8, 20, 24\}$
1	$\{7, 8, 11\}$	$\{7, 11, 21\}$

Check: $2 + 8 \times 3 = 26 = |GF(27)^*|$. ✓

Physical Identification [K]

Massive sector: fixed points = $\mathbb{Z}_2$

The two fixed points $\{+1, -1\}$ form the massive sector: every massive particle has an antiparticle ($\mathbb{Z}_2$ structure), consistent with the compact torus sector.

Massless $SU(3)$ sector: eight gluons

$$8 \text{ gluons} = 4 \mathbb{Z}_{13}\text{-orbits} \times 2 \mathbb{Z}_2\text{-parities} = |GF(9)^*|.$$

The three elements of each $\mathbb{Z}_{13}$ orbit correspond to the three colours $\{R, G, B\}$; the $\mathbb{Z}_2$ parity distinguishes colour from anticolour. This is the adjoint representation of $SU(3)$, derived from Galois structure alone.

$U(1)$ sector: photon from the fixed field $GF(3)$

The fixed field of the Frobenius on $GF(27)$ is $GF(3)$ (Fermat: $x^3 = x$ for $x \in GF(3)$). The projection $GF(27) \rightarrow GF(3)$ gives:

Structure	Sector
$GF(3)^* = \{1, 2\} \cong \mathbb{Z}_2$	Photon: $U(1)$, own antiparticle
$27 - 3 = 24$ remaining elements	$SU(3)$ sector (quarks, gluons)

Algebraic Necessity [B]

$$\text{Non-fixed elements in } GF(27)^* = 26 - 2 = 24, \quad (1)$$

$$24 = 8 \cdot 3 = |GF(9)^*| \cdot |\text{colours}| = (3^2 - 1) \cdot 3 = 3^3 - 3, \quad (2)$$

$$8 \text{ gluons} = \frac{26-2}{3} = 8. \quad (3)$$

All three relations follow solely from $|GF(27)| = 3^3$ and the order 3 of the Frobenius on $\mathbb{Z}_{13}$.

Complete Picture

Galois structure	Physical sector	Symmetry
$GF(3)^* = \mathbb{Z}_2$	Photon	$U(1)$
Fixed points $\{+1, -1\} \subset GF(27)^*$	Massive particle/antiparticle	$\mathbb{Z}_2$
4×2 three-orbits in $GF(27)^*$	8 gluons	$SU(3)_{\text{adj}}$

The relation $8 \times 3 = 24 = \text{Kissing}(D_4)$ is algebraically forced ($3^3 - 3 = 24$). Whether $\text{Kissing}(D_4)$ has a deeper geometric connection to this Galois structure is subject to further investigation.

Verification script: `pruef_339_frobenius.py` (6/6 assertions).

Declared 29 August 2026